Year 13 Maths - Further applications of calculus

Similar questions - Connected rates and parametric area

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A funnel is in the shape of an inverted cone with semi-vertical angle 45° and height 4 cm. Initially it is full. Liquid drains from a hole at the vertex so that $\dfrac{\mathrm{d}V}{\mathrm{d}t} = -3\sqrt{h}$, where $V\,\mathrm{cm}^3$ is the volume of liquid remaining and h cm is its depth after t seconds.

(a) Show that $V = \frac{1}{3}\pi h^3$. **[2]**

(b) Show that $\dfrac{\mathrm{d}h}{\mathrm{d}t} = -\dfrac{3}{\pi}h^{-3/2}$. **[3]**

(c) Find the exact time taken for the funnel to empty. **[4]**

2 A curve has parametric equations $x = \dfrac{3}{t+1}$, $y = 2t+1$ for $t \geqslant 0$. The region R is bounded by the curve, the x-axis, and the lines $x = 1$ and $x = 3$.

(a) Find the values of t at which the curve meets the lines $x = 3$ and $x = 1$. **[2]**

(b) Show that the area of R is given by $\displaystyle\int_0^2 \frac{3(2t+1)}{(t+1)^2}\,\mathrm{d}t$. **[3]**

(c) Hence show that the area of R is $6\ln 3 - 2$. **[4]**

Question 1

(a) Semi-vertical angle $45°$ so radius $r = h$, and $V = \frac{1}{3}\pi h^2 \cdot h = \frac{1}{3}\pi h^3$

(b) $\dfrac{\mathrm{d}V}{\mathrm{d}h} = \pi h^2$, so $\dfrac{\mathrm{d}h}{\mathrm{d}t} = \dfrac{-3\sqrt{h}}{\pi h^2} = -\dfrac{3}{\pi}h^{-3/2}$

(c) $t = \displaystyle\int_4^0 -\frac{\pi}{3}h^{3/2}\,\mathrm{d}h = \frac{\pi}{3}\left[\tfrac{2}{5}h^{5/2}\right]_0^4 = \frac{\pi}{3}\cdot\frac{64}{5} = \frac{64\pi}{15}$ seconds (≈ 13.4 s)

Question 2

(a) $x = 3$ at $t = 0$; $x = 1$ at $t = 2$

(b) Area $= \displaystyle\int_1^3 y\,\mathrm{d}x = \int_{t=2}^{t=0} y\frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t$ with $\dfrac{\mathrm{d}x}{\mathrm{d}t} = -\dfrac{3}{(t+1)^2}$. Reversing the limits removes the minus sign: $\displaystyle\int_0^2 \frac{3(2t+1)}{(t+1)^2}\mathrm{d}t$

(c) $\dfrac{2t+1}{(t+1)^2} = \dfrac{2}{t+1} - \dfrac{1}{(t+1)^2}$, so area $= 3\left[2\ln(t+1) + \dfrac{1}{t+1}\right]_0^2 = 3\left(2\ln 3 + \tfrac{1}{3} - 1\right) = 6\ln 3 - 2$